

BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

Provide Executive Summary of your project

Consilium is an AI Decision Copilot purpose-built for solo and small-team veterinary clinics in Southeast Asia. The platform compresses the three pain-points of a working vet's day — preparing for the next consult, capturing structured notes during it, and following up with owners after discharge — into a single decision layer powered by Claude Haiku 4.5 and Sonnet 4.6, Deepgram speech-to-text, and a Telegram-based owner channel that escalates only the cases that genuinely need a doctor's eyes.

The Southeast Asian companion-animal market is growing at ~9% CAGR, but solo vets still spend ~2.5 hours per night on after-hours admin (SOAP notes, owner follow-ups, billing reconciliation). Existing practice-management software (ezyVet, Provet Cloud) is a record-keeping system, not a decision layer — it does not write SOAP notes, does not triage owner messages, and does not produce a portable patient record. Consilium owns that decision layer end-to-end.

*The business model is a flat **RM 299/clinic/month SaaS** with a viral acquisition wedge: every closed case ships the owner a QR-scannable **Pet Health Passport, and every other clinic that scans that passport becomes our next inbound lead. We are raising RM 1.5M seed** to convert 3 pilot clinics (~80 consults/week, 220 follow-up triages over the last 30 days) into a 50-clinic Malaysian footprint within 12 months, then expand into Singapore, Indonesia, Thailand and the Philippines.*

2. Product Overview / Solution

Consilium is a single Next.js 16 web application plus a grammarY-powered Telegram bot. The app surfaces six features (F1–F6) that map onto the working vet's day, all backed by Anthropic Claude tool-use loops with strict structured-output guardrails.

Key Features: *Highlight key features of your product*

- **F1 — Pre-consult brief (/consult).* Claude Haiku 4.5 reads the patient's prior visit history, vaccinations, allergies and the receptionist's chief-complaint, and emits a 5-line briefing card the doctor can scan in 10 seconds before opening the chart.*
- **F2 — Voice-first consult capture (/consult).* Deepgram nova-3 transcribes the consult dictation in Bahasa Malaysia, English and Mandarin code-switching. Sonnet 4.6 then runs a parallel four-agent pipeline (SOAP / Rx / Billing / Todos) with tavily_search for drug-recall and fresh-guidance lookups, plus multimodal vision for wound photos, lab images and X-rays.*
- **F3 — Telegram follow-up triage (@consilium_vet_bot).* A grammarY bot polls inbound owner replies. Sonnet 4.6 runs a two-turn tool-use loop (request_photo, request_temperature, request_appetite_timeline, request_medication_compliance, schedule_doctor_callback) and*

emits a terminal decision: clear, monitor or escalate. Owner photos pass to Claude vision and join the differential.

- **F4 — Realtime doctor dashboard (/dashboard). * Supabase Realtime pushes any escalate decision to the dashboard within ~1.2 seconds. The doctor never has to refresh; escalations surface as cards on top of the queue.*
- **F5 — Doctor feedback loop (/api/corrections). * Every doctor edit on a generated SOAP note, prescription, or triage decision is persisted to a corrections table and consolidated nightly into clinic-specific SOPs stored in a LangGraph store (see brain.md). The next AI call inherits the doctor's previously-validated judgments as few-shot examples — the model gets sharper for that specific clinic over time.*
- **F6 — QR-scannable Pet Health Passport (/passport). * On case-close the clinic ships the owner a Telegram message + a public passport URL with a scannable QR. Any other clinic that scans the QR sees the patient's full history — and becomes our next inbound lead.*

Technology Used: Highlight the technology used in your application

- **App framework. * Next.js 16 (App Router) + React 19 + Tailwind CSS v4. Single repository, single deploy target (Vercel free tier today).*
- **Reasoning + vision. * Anthropic Claude — Haiku 4.5 (low-latency brief, ~2.4 s median live), Sonnet 4.6 (consult and triage). Per-feature model overrides via env vars (ANTHROPIC_MODEL_BRIEF, ANTHROPIC_MODEL_CONSULT, ANTHROPIC_MODEL_TRIAGE).*
- **Speech-to-text. * Deepgram nova-3, batch-mode in finals demo, WebSocket streaming for production.*
- **Web-search tool-use. * Tavily, 7-day cached in tavily_cache. The model only calls it when drug-recall or fresh-guidance is required; results stream into the consult UI as a banner.*
- **Database + Realtime + Storage. * Supabase (Postgres + Realtime channels + public Storage buckets consult-photos / owner-photos).*
- **Telegram bot. * grammY (TypeScript), polling for dev, webhook-ready route at app/api/telegram/webhook for production.*
- **Clinic-specific Brain (post-finals). * A LangGraph Python sidecar (agent/) with PostgresSaver for short-term memory and AsyncPostgresStore for synthesized clinic SOPs. Nightly cron consolidates raw visits + corrections into clean rules.*
- **Deployment. * Vercel for the web app; Telegram bot runs as a deployed webhook in production. Supabase managed Postgres for data.*

3. Market Research & Market Dynamics

3.1 Problem Landscape

*Solo and small-team veterinary clinics across Southeast Asia are drowning in administrative load. A typical Malaysian vet sees 20–30 consults a day, then spends roughly *2.5 hours every evening* writing SOAP notes, reconciling billing line items against the clinic's price list, and replying to anxious owners on WhatsApp who want to know if a post-op pet is recovering normally. The owner-follow-up channel is the worst of these: messages arrive at 11 p.m., they are mostly low-acuity ("she ate her breakfast"), and the few that genuinely need urgent attention ("there's blood at the incision and she won't stand") are buried in the noise. Practice-management software like ezyVet and Provet Cloud exists, but it is a passive record of what the doctor already wrote — it does not draft SOAP notes from a voice dictation, does not triage WhatsApp messages, and does not produce a portable record the owner can take with them. The result: real clinical decisions get made on a phone screen at midnight, and the patient records fragments the moment the owner switches clinics.*

3.2 Market Niche

*Consilium targets *solo and small-team general-practice vet clinics in tier-1 and tier-2 SEA cities* — clinics with 1–4 vets, RM 30K–150K monthly revenue, an existing WhatsApp/Telegram owner channel, and no in-house IT team. These clinics value (a) reducing nightly admin, (b) fewer "did you mean..." follow-up messages, and (c) a defensible record that follows the patient between providers. On the supply side, our partner clinics are the local equivalent of an SME — they cannot afford a RM 50K/year enterprise PMS, but they will absolutely pay RM 299/month for a tool that gives them their evenings back.*

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	All independent and small-chain veterinary clinics across Southeast Asia (MY, SG, ID, TH, PH, VN); ~38,000 clinics × RM 3,588/year	~RM 136 million ARR

SAM	Tier-1 and Tier-2 city clinics in Malaysia, Singapore, and Thailand (~12,500 clinics)	~RM 45 million ARR
SOM	Initial 18-month capture in Malaysia — ~600 clinics at RM 299/month + ancillary revenue	~RM 2.6 million ARR

Number-of-clinics estimates triangulate Malaysian Veterinary Council registrations, ASEAN companion-animal industry reports and clinic-density observations from our 3 pilot partners.

3.4 Market Dynamics & Placement:

Three concurrent forces make 2026 the right moment for Consilium:

- **Companion-animal spending is structurally rising.* SEA pet-care spending is growing ~9% CAGR through 2030, and the marginal owner is younger, urban, and treats the pet as a family member — meaning higher willingness to pay for clinical quality and faster, clearer post-op communication. This pushes clinics to upgrade their owner-facing experience or lose the customer to the next clinic over.*
- **Generative-AI cost has crossed the threshold for clinic economics.* A full Consilium loop (brief + consult + triage + Tavily) costs roughly RM 0.40–0.80 in Anthropic + Deepgram + Tavily token spend. At a flat RM 299/month subscription and ~80 consults/week, our gross margin sits comfortably above 80% even before prompt-cache discounts.*
- **Owner-side messaging is already on Telegram and WhatsApp.* Clinics do not need to onboard owners onto a new app — Consilium's bot rides on the channel owners already use. This eliminates the largest historic barrier to vet-software adoption (owner re-onboarding) and lets a clinic activate Consilium in a single visit.*

4. Value Proposition & Competitive Advantage

*Consilium is not another vet PMS. It is an ***AI decision layer*** that sits on top of (and eventually replaces) the parts of a PMS the doctor actually touches: the SOAP note, the prescription line items, the post-op follow-up message. We compete on three structural moats that an incumbent PMS cannot copy without a ground-up rebuild:*

- ****A correction-trained clinic brain.*** Every doctor edit feeds the corrections table, which is consolidated nightly into clinic-specific SOPs in a LangGraph store. Within 30 days a clinic's Consilium instance behaves like that clinic's head doctor — preferred drug choices, preferred billing codes, escalation thresholds. This is a per-clinic moat that compounds with usage and is invisible to competitors training on generic veterinary corpora.*
- ****The QR-scannable Pet Health Passport as a viral acquisition wedge.*** Every closed case ships a public passport URL the owner can scan at any other clinic. Each scan at a non-Consilium clinic is a warm inbound lead — an SOP that a paying customer has handed us. No incumbent PMS owns the patient-portable record because their data model is clinic-locked.*
- ****Voice-first, multilingual, multimodal capture with strict structured-output guardrails.*** Deepgram nova-3 handles Bahasa / English / Mandarin code-switching on a single dictation; Claude Sonnet 4.6 emits SOAP/Rx/Billing through emit_consult schema validation and routes wound photos, lab images and X-rays through vision in the same tool-use loop. Competing tools either ship in English-only, or ship without structured-output guardrails and hallucinate prescriptions.*

These compound: the more consults a clinic runs through Consilium, the better the brain becomes for that clinic; the more passports the clinic ships, the more inbound scans we receive at other clinics; and every inbound clinic that signs up lifts the brain across the network for shared SOPs (drug recalls, regional outbreaks).

5. Competitor Analysis (Porter's Five Forces)

<i>Force</i>	<i>Intensity</i>	<i>Assessment</i>
Competitive Rivalry	<i>Medium</i>	<i>PMS players (ezyVet, Provet Cloud, VetPanel) focus on record-keeping, not AI decision-making. AI scribes target human medicine, not veterinary workflows.</i>
New Entrants	<i>Medium–High</i>	<i>AI tools are easier to build, but defensibility comes from clinic-specific data and patient passport built over time.</i>
Buyer Power	<i>Low–Medium</i>	<i>Clinics are price-sensitive, but switching cost increases once workflows and data are embedded.</i>
Supplier Power	<i>Medium</i>	<i>Dependence on AI providers (Anthropic, Deepgram, Tavily), mitigated through abstraction and cost control.</i>
Substitutes	<i>Low</i>	<i>Current alternative is manual workflows, which are inefficient and not scalable.</i>

Competitor Comparison Table:

<i>Competitor</i>	<i>Strength</i>	<i>Weakness</i>	<i>Consilium Advantage</i>
ezyVet / Provet Cloud	Strong PMS, billing systems	No AI, no triage, expensive	Adds decision layer at lower cost
Heidi / Suki / Abridge	Advanced AI scribes	Built for human medicine	Veterinary-specific + owner triage
VetPanel (SEA)	Local PMS presence	No AI layer	Integrates as AI layer above PMS
WhatsApp / Paper	Familiar, free	No structure, no automation	Replaces entire workflow with structured system

6. Business Model Canvas

6.1. Customer Segmentation

1. - *Solo and small-team vet clinics (primary).* 1–4 vets, urban or peri-urban tier-1/tier-2 SEA cities, RM 30K–150K monthly revenue, existing WhatsApp/Telegram owner channel. They pay the RM 299/clinic/month subscription.
2. - *Pet owners (channel users, not paying customers).* They interact with Consilium through the Telegram bot and the passport URL. They never see a paywall — the clinic pays for them.
3. - *Multi-branch chains (expansion segment).* Once we have a 50-clinic Malaysian footprint, regional chains (e.g. The Pet Vet Group, Animal Medical Centre franchises) become a natural land-and-expand target with a multi-seat enterprise tier.

6.2. Revenue Stream

Primary — Flat clinic subscription. RM 299/clinic/month, billed annually. Includes unlimited consults, unlimited triage, unlimited Telegram-bound owners, unlimited passports.

Secondary — Usage-based and ancillary.

- - *Per-consult overage* (above 200 consults/month): RM 1.50/consult — protects margin on power-user clinics without capping the base plan.

- - *Multi-branch enterprise tier:* RM 199/clinic/month for chains with 5+ branches, plus a one-time RM 5,000 onboarding fee for SOP migration and brain setup.
- - *Verified-vendor billing-matrix expansion:* Pharmaceutical and supplement vendors pay to have their SKUs verified into the diagnosis-keyed billing matrix. Sponsored line items are clearly flagged in the UI; the doctor remains the decision-maker. RM 1,200–4,800/SKU/year.
- - *API access (post-finals):* Practice-management software vendors (ezyVet, Provet, VetPanel) pay a RM 0.20/call API tier to embed Consilium's scribe and triage into their own UI as a white-label layer.

At 600 paying clinics × RM 299/month, base ARR is *~RM 2.15 million. Realistic blended ARPC including overage and enterprise lift is ~RM 360/month → *~RM 2.6M ARR by month 18**.

6.3. Cost Structure

Cost Driver	Cost Type
• Engineering payroll (3 FTE founders + 2 hires by month 12)	Fixed
• Anthropic Claude API (Haiku 4.5 + Sonnet 4.6, prompt cache)	Variable (~RM 0.40–0.80 per closed case)
• Deepgram nova-3 transcription	Variable (~RM 0.06 per voice-minute)
Tavily search (7-day cached)	Variable (~RM 0.02 per cache miss)
Supabase (Postgres + Realtime + Storage)	Fixed (Pro tier)
Vercel hosting	Fixed (Pro tier)
Clinic onboarding + SOP setup (pilot phase)	Variable (founder time)
Compliance, legal, licensing	Overhead

6.4. Key Partners

Internal	External
Anthropic Claude (reasoning + vision)	Malaysian Veterinary Council (MVC) — clinic verification
Deepgram (speech-to-text)	Pilot clinics (3 currently, target 50 by month 12)
Supabase (database + realtime + storage)	Pharmaceutical vendors (Zoetis, Boehringer Ingelheim, Elanco)
Tavily (search + drug recall)	Telegram (owner communication channel)
Vercel (hosting)	Regional vet schools (UPM, Chulalongkorn, Universitas Gadjah Mada)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
AI hallucination (wrong prescription dose)	High	Low–Medium	Doctor-in-the-loop approval, schema validation, clinic-specific correction feedback
API outage / rate limits	Medium	Low	Provider abstraction, fallback models, mock mode, prompt caching (~50% load reduction)
Owner privacy / consent issues	High	Medium	Consent at onboarding, secure UUID-based passport, signed access links
Regulatory changes (AI in healthcare)	Medium	Medium	Full audit logs, compliance roadmap toward HIPAA-equivalent standards
PMS competitors adding AI	High	Medium	Differentiation via clinic-specific brain, pet passport, pricing model
Clinic churn / adoption resistance	Medium	Medium	White-glove onboarding, workflow tuning, annual billing commitment

7. Implementation & Financials

7.1 Go-to-Market Strategy

We adopt a *city-by-city land-and-expand* play, identical in shape to e-hailing rollouts but at the SME scale of a vet clinic. Phase 1 (Klang Valley) focuses on 50 clinics in Kuala Lumpur, Petaling Jaya, Subang and Shah Alam — geographically dense enough that one founder can do white-glove onboarding in person within a 30-minute drive. Phase 2 expands to Penang and Johor Bahru once Klang Valley is at 25 clinics paying. Phase 3 enters Singapore (different regulatory regime, higher willingness-to-pay) and Bangkok (volume play).

The acquisition wedge is the *QR pet passport. Every passport scanned at a non-Consilium clinic is a warm inbound lead — we cold-call those clinics with "your patient X just walked in carrying our passport, here's a 30-day free trial." This is materially cheaper than paid acquisition (estimated CAC: **RM 150–250/clinic* vs. industry benchmark of RM 800–1,500 for veterinary SaaS).

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch (Klang Valley)	Pilot clinics → initial paying users; refine clinic-specific AI workflows; deploy secure passport sharing	Month 1–6	25 clinics
Expansion (Penang, JB)	Hire Customer Success; launch multi-clinic features; improve AI workflow reliability	Month 6–12	200 clinics across 3 cities
SEA Expansion + Optimisation	Enter SG & Bangkok; introduce enterprise/API offerings; optimise revenue streams	Month 12–18	600 clinics, ~RM 2.6M ARR
Total Outcome	Scaled deployment with strong margins and sustainable growth	18 months	Multi-country presence

7.3 Revenue Projections (e.g 3-Year)

Metric	Year 1	Year 2	Year 3
<i>Active Cities</i>	3	6	10
<i>Paying Clinics</i>	200	800	2,200
<i>Avg Revenue / Clinic (RM/month)</i>	320	360	410
<i>Subscription Revenue</i>	RM 0.77M	RM 3.46M	RM 10.82M
<i>Additional Revenue (API / Enterprise)</i>	RM 0.12M	RM 0.84M	RM 3.50M
Total Revenue	RM 0.89M	RM 4.30M	RM 14.32M
<i>Net Position</i>	(RM 1.10M)	(RM 0.20M)	RM 4.20M

7.4 Key Financial Milestones

- ***Month 6:** First 25 paying clinics in Klang Valley. Per-clinic brain consolidation cron in production. Time-limited signed passport URLs shipped.
- ***Month 12:** 200 paying clinics across KL, Penang, JB. LangGraph sidecar in production path. First 5 enterprise multi-branch contracts signed.
- ***Month 18:** 600 paying clinics across MY, SG, TH. Reached ***break-even contribution margin per clinic***. RM 2.6M ARR. Series A conversation opened.
- ***Month 30:** Cashflow-positive at clinic level; raise Series A (RM 12–15M) for Indonesia, Vietnam and Philippines expansion plus one enterprise PMS-integration partnership.